MyTrip System Structure and Algorithms Report

MyTrip – Budget Road Trip Planner

CS 4398 – Software Engineering Project

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1. Project Overview

MyTrip – Budget Road Trip Planner is a web-based application created to help travelers organize a road trip and understand its likely cost before leaving. The system addresses a common planning problem: route information, fuel cost, lodging, activities, and budget decisions are often handled in separate places. MyTrip brings these planning tasks into one academic prototype.

The target users are travelers who want a straightforward way to build and save a trip. A traveler signs in with a demonstration account, enters a starting location, an optional waypoint, a destination, route mileage, a target budget, and vehicle information. The application displays the route with Google Maps, estimates fuel cost and drive time, presents simulated lodging, fuel-station, and attraction choices, and summarizes the expected trip cost.

The main workflow is to plan a route, review route options, select relevant trip items, compare the calculated total with the budget, and save the trip under the traveler's email. A saved trip can later be reopened or deleted. The current integration version demonstrates this primary workflow, although the Edit Trip process is not fully functional.

2. System Architecture

MyTrip uses a layered Spring Boot architecture. Separating presentation, request handling, business logic, data access, and persistence makes the system easier to understand, test, and maintain.

Presentation layer: HTML, CSS, and JavaScript collect input and display route, selection, account, and summary information in the browser.

Controller layer: TripController receives HTTP or REST requests and returns application responses.

Service layer: TripService coordinates validation, calculations, ownership rules, trip lookup, saving, and deletion.

Repository layer: TripRepository provides Spring Data JPA access to stored Trip records.

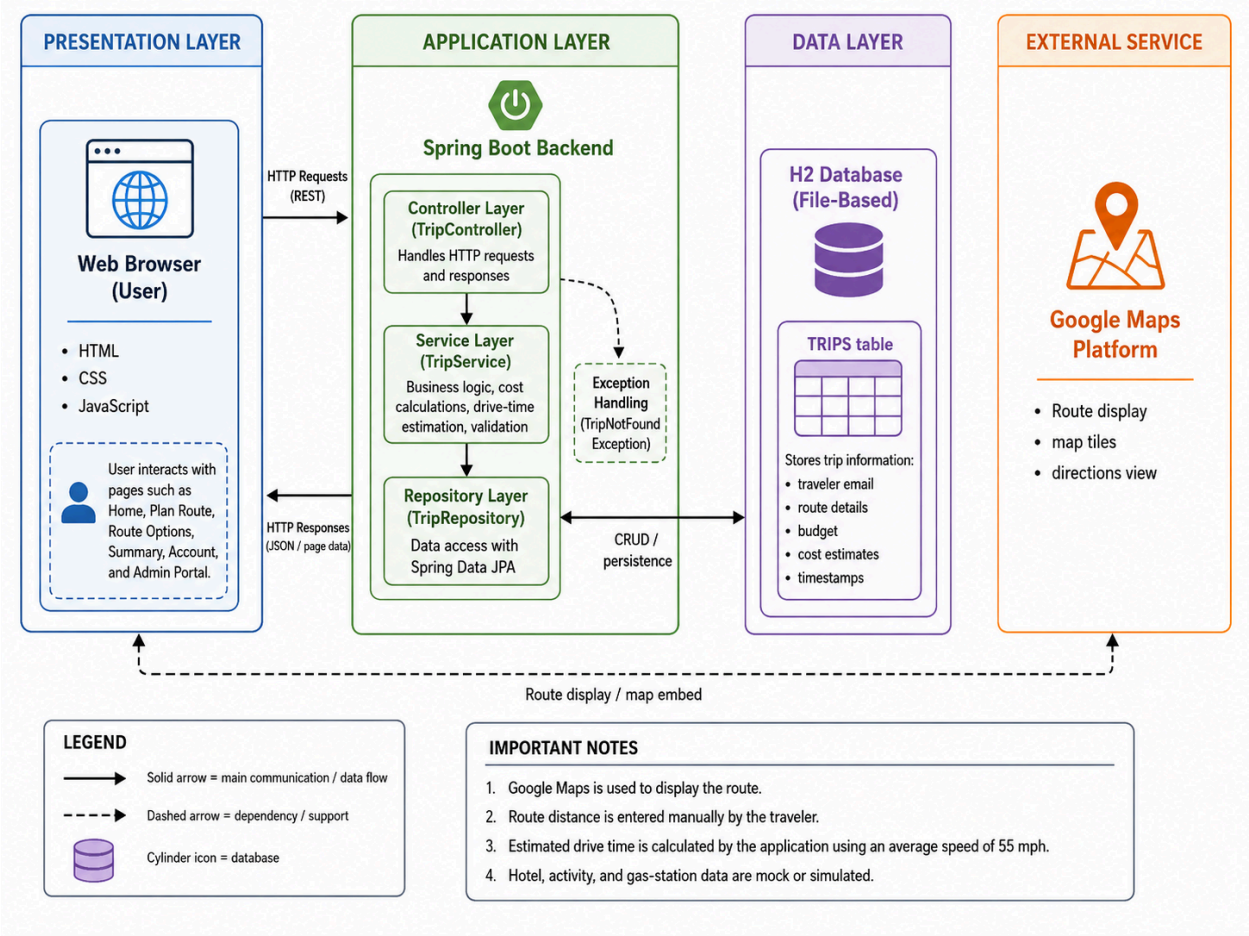
Data layer: A file-based H2 database stores saved trip information locally.

External route display: Google Maps displays the route and gives the traveler a visual mileage reference.

The normal backend request flow is:

Web Browser → TripController → TripService → TripRepository → H2 Database

Responses travel back through the backend and are returned to the browser as application data. TripController does not communicate directly with the database; repository access is coordinated through TripService.



3. Package Structure

The backend is organized under the base package com.mytrip. An appropriate Spring Boot organization separates classes by responsibility while keeping related behavior together.

The application entry point contains MytripApplication and starts the Spring Boot application.

The controller area contains request-handling logic such as TripController.

The service area contains business logic such as TripService.

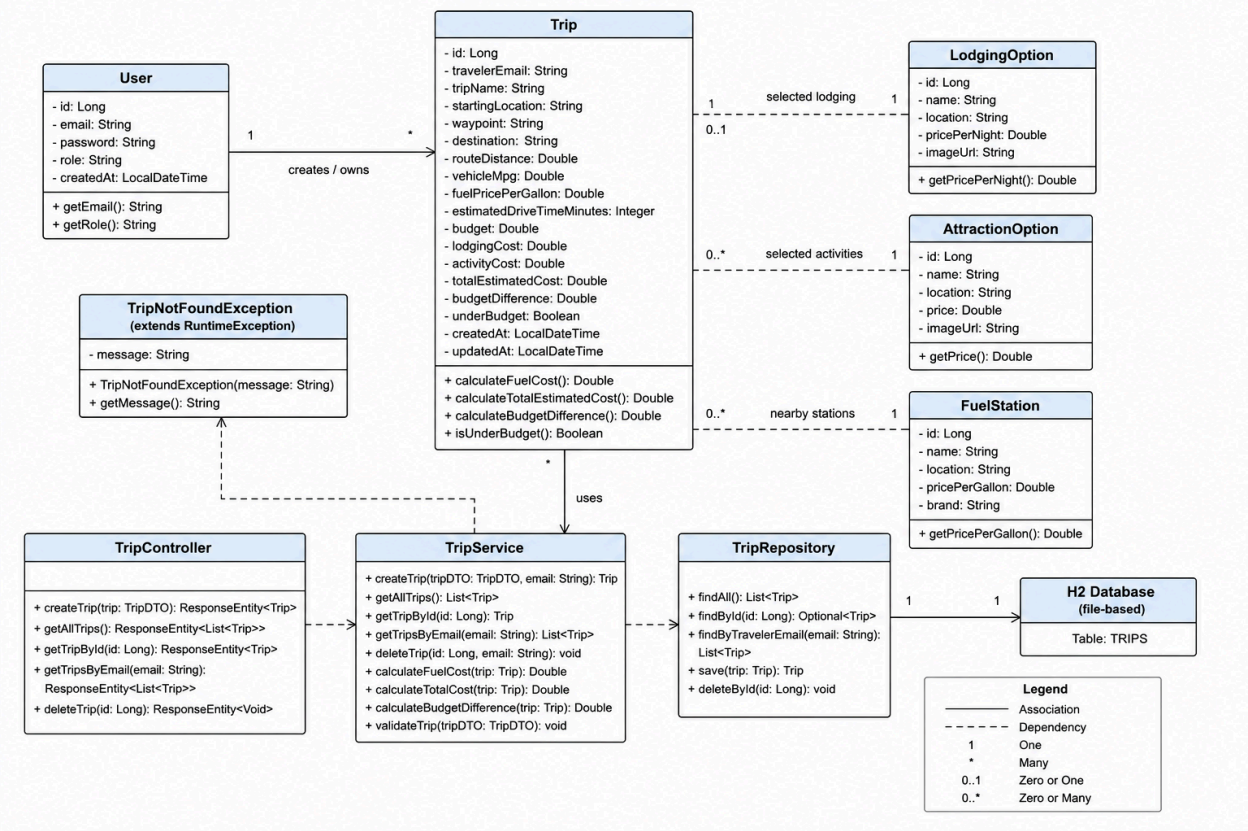
The repository area contains data-access interfaces such as TripRepository.

The model or entity area contains the Trip entity.

The exception-handling area contains application-specific errors such as TripNotFoundException.

Frontend static resources contain the HTML pages, CSS styles, JavaScript behavior, and related assets.

This organization supports clear responsibilities without requiring the frontend, controller, service, and repository code to be combined in the same package.



4. Model Layer

The Trip entity represents a saved road-trip plan. It carries the information needed to identify the trip, perform cost calculations, associate the record with its owner, and store the result in H2.

Representative trip data includes a trip ID, traveler email, trip name, starting location, optional waypoint, destination, route distance, vehicle MPG, fuel price per gallon, estimated drive time, budget, lodging cost, activity cost, total estimated cost, budget difference, under-budget status, and created and updated timestamps.

The trip ID distinguishes one saved record from another. Traveler email supports ownership and filtering. Numeric fields provide inputs and outputs for the algorithms, while the Boolean under-budget value records whether the final estimate fits the target budget. Timestamp values support record history and allow the system to identify when a trip was created or changed.

5. Controller Layer

TripController is responsible for receiving HTTP or REST requests from the frontend and communicating with TripService. It acts as the boundary between browser requests and backend application logic.

Controller operations support creating or saving trips, retrieving all trips when appropriate, retrieving a trip by ID, retrieving trips by traveler email, reopening stored trips, deleting trips, and returning validation or error responses. The controller delegates calculations, ownership checks, repository access, and other business rules to TripService rather than accessing H2 directly.

6. Service Layer

TripService contains the central business logic for the application. It coordinates data received from TripController, validates important values, performs trip calculations, applies traveler ownership rules, and uses TripRepository for persistence.

- calculating gallons required and estimated fuel cost;
- adding fuel, lodging, and selected activity costs;
- comparing the estimate with the target budget;
- associating saved trips with a traveler email;
- finding trips by ID or owner;
- saving and deleting trip records; and
- raising or handling exceptions when data is invalid or a trip cannot be found.

Keeping these rules in the service layer helps the same behavior remain consistent across controller operations and prevents calculation logic from being spread across the frontend and repository.

7. Repository and H2 Database

TripRepository is the Spring Data JPA data-access interface for Trip records. Spring Data JPA supplies standard create, read, and delete behavior and supports lookup of trips associated with a traveler email.

MyTrip uses file-based local H2 persistence with the following connection location:

jdbc:h2:file:./data/mytrip

Because the database is file-based, saved trips remain after a browser page refresh and can remain after an application restart. However, each local installation uses its own database file. A trip saved on one team member's computer is not automatically shared with another computer.

The H2 console can be accessed in the local development environment. Stored trip records can be inspected with:

```
SELECT * FROM TRIPS;
```

8. Frontend Structure

The presentation layer uses HTML for page structure, CSS for layout and visual styling, and JavaScript for interaction, validation, calculation display, selections, and communication with the backend.

Home: introduces MyTrip and provides access to the main planning workflow.

Plan Route: collects the starting location, optional waypoint, destination, budget, mileage, vehicle, and traveler information.

Route Options: displays the Google Maps route, manually entered mileage, estimated drive time, and available planning options.

Summary: shows fuel, lodging, activity, total, budget, and group-splitting results.

Account: shows trips saved under the signed-in traveler's email and supports reopening or deleting them.

Admin Portal: provides demonstration-level administrative access for the academic prototype.

The browser displays external Google Maps route information but uses mock lodging, attraction, and fuel-station listings for the prototype. JavaScript sends relevant trip requests to the Spring Boot backend and displays the returned application data.

9. Fuel-Cost Algorithm

The fuel-cost calculation uses route distance, vehicle efficiency, and fuel price. Route distance and MPG must both be greater than zero.

$$\text{Gallons Required} = \text{Route Distance} \div \text{Vehicle MPG}$$

$$\text{Estimated Fuel Cost} = \text{Gallons Required} \times \text{Fuel Price Per Gallon}$$

For the tested example, the route distance is 81.9 miles, the vehicle efficiency is 28 MPG, and the fuel price is approximately $3.38 per gallon.

$$81.9 \div 28 = 2.925 \text{ gallons}$$

$$2.925 \times \$3.38 \approx \$9.89$$

The estimated fuel cost is therefore approximately $9.89. Minor display differences can occur if the demonstration fuel price changes or if intermediate values are rounded differently.

10. Total-Cost and Budget Algorithms

Selected activity prices are added together before the final estimate is calculated.

$$\text{Activity Cost} = \Sigma \text{ Selected Activity Price}_i, \text{ for } i = 1 \text{ to } n$$

$$\text{Total Estimated Cost} = \text{Fuel Cost} + \text{Lodging Cost} + \text{Activity Cost}$$

$$\text{Budget Difference} = \text{Budget} - \text{Total Estimated Cost}$$

If the budget difference is zero or positive, the trip is within budget. If the value is negative, the trip is over budget. The summary also supports group cost splitting by dividing the total estimated cost among the number of travelers. The traveler count must be valid before a per-person amount can be calculated.

11. Drive-Time Estimation

MyTrip estimates drive time from the manually entered route mileage and an assumed average driving speed. It does not retrieve live travel time directly from Google Maps.

Estimated Drive Time = Route Distance ÷ Average Speed

Using the tested values of 81.9 miles and an average speed of 55 MPH:

81.9 ÷ 55 ≈ 1.489 hours ≈ 1 hour 29 minutes

The estimated drive time is displayed by MyTrip under the map on the Route Options page. It is a planning estimate and does not account for current traffic, construction, rest stops, or changing road conditions.

12. CRUD Operations

The current backend and interface support the following data operations:

- Create or save a trip.
- Read all trips when appropriate.
- Read one trip by its ID.
- Read trips filtered by traveler email.
- Delete a saved trip.

Reopening a saved trip restores its stored information so the traveler can review the plan. The Edit Trip workflow appears in the interface, but it is not fully functional in the current integration version. Update behavior should therefore not be treated as a successfully completed feature.

13. Trip Ownership by Traveler Email

Each saved Trip is associated with the traveler's email address. The account view requests and displays records filtered by that email, which prevents one demonstration traveler from seeing trips saved under another traveler's account.

The demonstrated accounts are traveler@example.com and traveler2@example.com. During acceptance testing, a trip saved under traveler@example.com did not appear when the application was opened under traveler2@example.com. This confirms the intended account-specific filtering behavior for the academic prototype.

This ownership approach supports separation of demonstration user data, but it does not replace production authentication and authorization. A production version would require secure identity verification and server-side access controls.

14. Validation and Exception Handling

Validation protects calculations and persistence from incomplete or invalid input. Important checks include route mileage greater than zero, budget greater than zero, valid MPG values, a required starting location, a required destination, and a signed-in traveler before saving.

Observed validation messages include:

> “Enter a valid route distance greater than zero.”
>
> “Error: Invalid target budget.”

Guest users are prevented from saving trips. TripNotFoundException is the custom exception used when a requested trip cannot be found. The application can translate validation failures and missing-trip errors into responses that the frontend can display to the traveler.

15. Data Structures

The primary data structure is the Trip object, which groups the fields and calculated values for one saved plan. Collections are used conceptually wherever the application must display or process more than one item.

- lists of saved trips;
- lists of lodging options;
- lists of attractions;
- lists of fuel stations;
- collections of selected options;
- numeric values for distances, prices, costs, budgets, and traveler counts; and
- a Boolean value for under-budget status.

Lists are appropriate because the number of saved trips or selectable options can vary. They also allow the frontend to repeat a consistent display component for each result. Numeric and Boolean values support direct calculations and clear budget-status decisions.

16. External and Simulated Data

External Data

Google Maps provides the visual route display and map information. The displayed route mileage serves as a reference, but the traveler manually enters that mileage into MyTrip. The application does not automatically retrieve route distance from Google Maps.

Simulated or Mock Data

lodging listings;

attraction listings;

fuel-station listings;

booking actions;

reservation confirmations;

payment behavior; and

changing demonstration gas-price values.

These simulated elements allow the team to demonstrate the complete planning experience without connecting to commercial reservation or payment services. MyTrip does not perform real hotel reservations, real activity bookings, or real payments.

17. Known Limitations

MyTrip is an academic prototype, and its limitations should be understood when reviewing the current implementation.

Manual route mileage: Route distance is not retrieved automatically. The traveler enters the mileage shown on the map.

Estimated drive time: Drive time uses an assumed average speed instead of live Google Maps travel time.

Mock travel content: Hotel, activity, and fuel-station information is simulated rather than retrieved from production providers.

Simulated transactions: Reservations, confirmations, and payments are demonstrations only.

Demonstration authentication: Login behavior is suitable for showing account-specific features but is not production-level security.

Incomplete editing: The Edit Trip workflow is not fully functional in the current integration version.

Local persistence: Each installation uses its own H2 database file, so data is not shared automatically across computers.

Minor interface issues: Small formatting or interface inconsistencies may remain in the prototype.

These limitations are acceptable for the current course prototype because the system demonstrates the main architecture, algorithms, persistence behavior, and traveler workflow. A future production version could add secure authentication, server-hosted shared persistence, automatic route-distance and traffic data, real provider integrations, transaction security, and a complete update workflow.

18. Conclusion

MyTrip combines an HTML, CSS, and JavaScript browser interface with a Spring Boot backend written for Java 21. TripController receives application requests, TripService coordinates validation and business rules, TripRepository provides Spring Data JPA persistence, and H2 stores saved Trip records locally. Google Maps supplies the route display, while MyTrip calculates fuel cost, total estimated cost, budget difference, drive time, and group cost splitting.

The system demonstrates the primary road-trip planning workflow: a traveler can sign in, create a route plan, enter route mileage, review simulated travel options, compare the estimate with a budget, save the trip, reopen it, and delete it. Email-based filtering separates saved trips between demonstration accounts.

The current version meets the main goals of an academic software engineering prototype while clearly leaving production authentication, real external booking and payment services, automatic mileage retrieval, shared deployment data, and complete trip editing for future improvement.